MATH 241: Calculus III

Fall 2026

Lecture 9: The Chain Rule for Multivariable Functions

Background & Motivation

In Calc I, the chain rule handled compositions like $f(g(x))$. Now our functions have multiple intermediate variables, each depending on one or more parameters. The multivariate chain rule keeps track of all the paths from input to output—and in matrix form, it becomes the Jacobian.

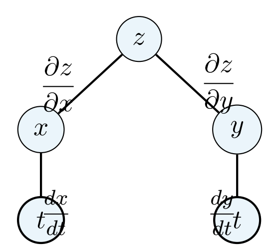
1. Chain Rule: One Parameter (Case 1)

Theorem 1: Chain Rule – Case 1: $z = f(x, y)$, $x = x(t)$, $y = y(t)$

If $z = f(x, y)$ where x and y are both functions of a single parameter t:

$$\frac{dz}{dt} = \underline{\hspace{6cm}}$$

– Each intermediate variable contributes one term: (partial w.r.t. that variable) $\times$ (derivative of that variable w.r.t. t)



Tree diagram for Case 1. z depends on x and y; each of x, y depends on t. To find $\frac{dz}{dt}$, walk every path from z down to t, multiply the derivatives along each path, and sum over all paths.

Example 1: $z = x^2 y$ with $x = t^2$, $y = \sin t$

Find $\dfrac{dz}{dt}$.

2. Chain Rule: Multiple Parameters (Case 2)

Theorem 2: Chain Rule – Case 2: $z = f(x, y)$, $x = x(u, v)$, $y = y(u, v)$

$$\frac{\partial z}{\partial u} = \underline{\hspace{6cm}}$$

$$\frac{\partial z}{\partial v} = \underline{\hspace{6cm}}$$

Matrix Form of the Chain Rule

$$\begin{pmatrix} \frac{\partial z}{\partial u} \\ \frac{\partial z}{\partial v} \end{pmatrix} = \begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} \end{pmatrix} \begin{pmatrix} \frac{\partial z}{\partial x} \\ \frac{\partial z}{\partial y} \end{pmatrix}$$

– The chain rule in matrix form is the Jacobian matrix

Example 2: $z = \ln(xy)$ with $x = e^u$, $y = \cos v$

Find $\frac{\partial z}{\partial u}$ and $\frac{\partial z}{\partial v}$.

3. Implicit Differentiation

Implicit Differentiation via Partial Derivatives

If $F(x, y) = 0$ defines y implicitly as a function of x:

$$\frac{dy}{dx} = \underline{\hspace{6cm}} \quad \text{(provided } F_y \neq 0\text{)}$$

If $F(x, y, z) = 0$ defines z implicitly as a function of x and y:

$$\frac{\partial z}{\partial x} = \underline{\hspace{3cm}}, \qquad \frac{\partial z}{\partial y} = \underline{\hspace{3cm}}$$

– *Why it works:* Differentiate $F(x, y(x)) = 0$ with respect to x using the chain rule: $F_x + F_y \frac{dy}{dx} = 0$, then solve

Example 3: Implicit Differentiation: $x^3 + y^3 = 6xy$

Find $\dfrac{dy}{dx}$.

4. Total Differentials

Definition 1: Total Differential

The **total differential** of $f(x, y)$:

$$df = \underline{\hspace{6cm}}$$

– This is the best linear approximation: $\Delta f \approx df$ for small $dx = \Delta x$, $dy = \Delta y$
For $f(x, y, z)$: $df = f_x\,dx + f_y\,dy + f_z\,dz$

Example 4: Temperature Estimation with Differentials

The temperature in a room is modeled by $T(x, y) = 100 - x^2 - 2y^2$ (degrees). A sensor moves from $(1, 2)$ to $(1.05, 2.1)$. Estimate the change in temperature.

Practice Problems

Problem 1. If $z = x^2 + y^2$, $x = t$, $y = e^t$, find $\dfrac{dz}{dt}$.

Problem 2. Let $w = xy + yz + xz, \quad x = u + v, \quad y = u - v, \quad z = uv$. Find $\dfrac{\partial w}{\partial u}$.

Problem 3. Find $\dfrac{dy}{dx}$ for the circle $x^2 + y^2 = 25$ using implicit differentiation with partials.

Problem 4. Use differentials to approximate $\sqrt{(3.02)^2 + (3.97)^2}$.

Problem 5. If $z = f(x - y, xy)$ where f is differentiable, express $x\dfrac{\partial z}{\partial x} + y\dfrac{\partial z}{\partial y}$ in terms of f_1 and f_2 (partials of f w.r.t. its first and second arguments).